Topic 6: Routing, Congestion Control, QoS & WANs

Course outline weeks 5-7: point-to-point networks, routing, congestion control, internetworking; plus SLA, SDN and Internet structure. Big-mark questions (6-10 marks).

How often this topic is examined

Question asked	Paper(s)	Marks
Sources of routing-table information	2023 Q2a	3
Link-state vs distance-vector (table)	2023 Q2b	6
Complete table: RIPv1 / EIGRP / OSPF - type & metric	2023 Q2c	6
How EIGRP works and selects the best path	2023 Q2d	5
How RIPv2 works	ISA2024 Q5c	6
Congestion control + two mechanisms	2025 Q3c	10
QoS: VoIP vs video; QoS parameters; 'best effort network'	2023 Q2e [5], Q3e [6], Q3d [4]	4-6
Network performance optimisation scenario	ISA2024 Q4d	7
Software-defined networking (SDN)	ISA2024 Q4b	5
Hierarchical structure of the Internet (ISPs)	2018 Q4c	6
SLA - purpose & contents	2023 Q5d	3
Two WAN protocols	SE2024 Q1c	2

1. Routing basics

Routing is the process by which a **router** selects the **best path** for a packet from the source network to the destination network, using the destination IP address and its **routing table** (network/prefix, next hop, outgoing interface, metric, administrative distance).

> **EXAM QUESTION** 2023 Q2a **[3]**
> Natalie has forgotten how routing tables work. Describe three different sources a router can use to put/maintain information in a routing table.

1. **Directly connected networks** - when an interface is configured with an IP address and is up, the router automatically adds that network (code **C** in Cisco).
2. **Static routes** - manually entered by the administrator (`ip route 10.0.0.0 255.0.0.0 192.168.1.2`), code **S**. Includes the **default route** (0.0.0.0/0) for 'everything else', e.g. towards the ISP.
3. **Dynamic routing protocols** - routers learn routes automatically by exchanging updates with neighbours using **RIP (R), OSPF (O), EIGRP (D)**, BGP; they adapt when the topology changes.

When several sources offer the same route, the router prefers the lowest **administrative distance**: connected 0, static 1, EIGRP 90, OSPF 110, RIP 120.

2. Distance vector vs link state

Made by King Tine, the AI Engineer

EXAM QUESTION 2023 Q2b **[6]**
Tabulate at least three main differences between link-state and distance-vector routing protocols.

Distance vector	Link state
Router knows only **distance (metric) and direction (next hop)** to each network - 'routing by rumour'	Every router builds a **complete map (topology) of the network** and calculates paths itself
Sends its **whole routing table periodically** (e.g. RIP every 30 s) to **directly connected neighbours** only	Sends small **Link-State Advertisements (LSAs)** to **all routers** (flooding) only **when a change occurs** (+ hellos)
Algorithm: **Bellman-Ford**	Algorithm: **Dijkstra's Shortest Path First (SPF)**
Slow convergence; prone to **routing loops** / count-to-infinity (needs split horizon, poison reverse, hold-down)	**Fast convergence**, loop-free
Simple, low CPU & memory	Needs more CPU and memory (LSDB, SPF calculations)
Suitable for small, simple networks	Scales to large hierarchical networks (areas)
Examples: **RIPv1, RIPv2**, IGRP (EIGRP = advanced DV)	Examples: **OSPF, IS-IS**

EXAM QUESTION 2023 Q2c **[6]**
Copy and complete the table: Routing protocol | Distance-vector or link-state? | Metric used.

Routing protocol	Distance-vector or Link-state?	Metric used
RIPv1	Distance-vector (classful)	**Hop count** (max 15; 16 = unreachable)
EIGRP	**Advanced distance-vector** (also called hybrid)	**Composite metric**: bandwidth and delay by default (can include load and reliability)
OSPF	Link-state	**Cost** - based on bandwidth (cost = reference bandwidth 100 Mbps / interface bandwidth)

3. EIGRP

EXAM QUESTION 2023 Q2d **[5]**
Describe the workings of the EIGRP routing protocol, clearly highlighting how it selects the best path for transmission of packets.

- **Enhanced Interior Gateway Routing Protocol** - Cisco-developed **advanced distance-vector** protocol (published as RFC 7868); classless (supports VLSM/CIDR); administrative distance 90.
- **Neighbour discovery:** sends **Hello** packets (multicast 224.0.0.10) to form adjacencies with neighbours; uses RTP (Reliable Transport Protocol) for reliable updates.
- **Three tables: neighbour table**, **topology table** (all routes learned from all neighbours), **routing table** (best routes).
- **Partial, bounded updates** - only sends changes, only to routers that need them (not periodic full tables) -> low bandwidth use.
- **Metric:** composite of **bandwidth (slowest link on the path) + delay (sum of delays)** by default; formula metric = 256 x (10^7 / slowest bandwidth in kbps + total delay in tens of microseconds). Lowest metric = best.

- **Best-path selection with DUAL (Diffusing Update Algorithm):** for every destination the route with the **lowest Feasible Distance (FD)** becomes the **Successor** and is put in the routing table. A backup route whose **Reported (Advertised) Distance < FD of the successor** (the *feasibility condition*) becomes the **Feasible Successor**, kept in the topology table.
- If the successor fails, the feasible successor is used **immediately** - very **fast convergence, loop-free**. If none exists, the router sends queries to neighbours.
- Supports equal and **unequal-cost load balancing** (variance command).

4. RIP version 2

EXAM QUESTION ISA2024 Q5c **[6]**
Explain in detail the working of the RIP v2 routing algorithm.

- **Routing Information Protocol v2** - a **distance-vector** protocol using the **Bellman-Ford** algorithm; metric = **hop count**; maximum **15 hops** (16 = unreachable) so suits small networks. AD 120, runs over **UDP port 520**.
- Each router starts with its directly connected networks, then **every 30 seconds** sends its **entire routing table** to neighbours using **multicast 224.0.0.9** (RIPv1 used broadcast).
- When a router receives an update it adds **1 hop** to each route; if the route is new or better (fewer hops) it updates its table with the neighbour as next hop.
- **Improvements over RIPv1: classless** - sends the **subnet mask** with each route (supports VLSM & CIDR); **authentication** (plain text / MD5); **next-hop** field; **route tags**; multicast updates.
- **Timers:** update 30 s; invalid 180 s (route marked unreachable if not heard); hold-down 180 s; flush 240 s (removed).
- **Loop prevention: split horizon** (don't advertise a route back out the interface it was learned on), **route poisoning** (advertise failed route with metric 16), **poison reverse**, **hold-down timers**, **triggered updates** (send immediately on change).
- Disadvantages: slow convergence, hop count ignores bandwidth, periodic updates waste bandwidth.

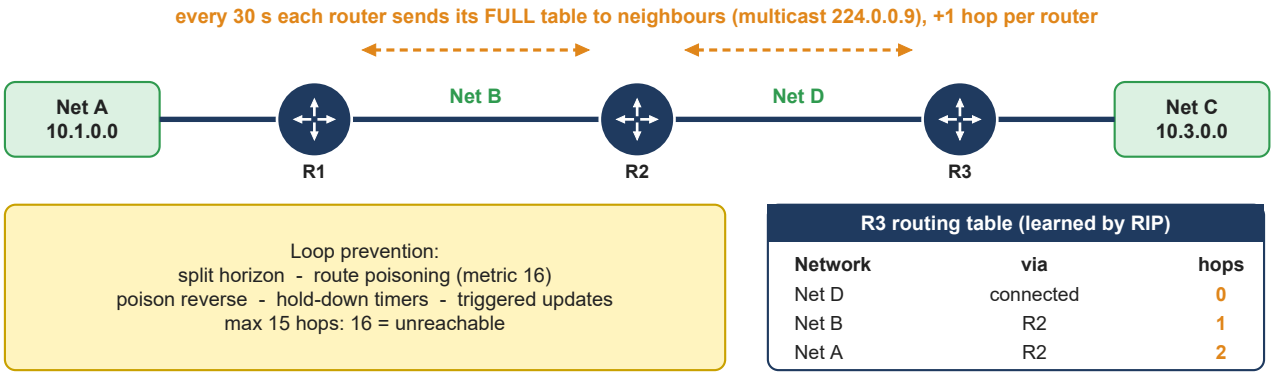


Figure: RIP: hop counts, updates and loop prevention

5. Congestion control

EXAM QUESTION 2025 Q3c **[10]**
A network is experiencing a high volume of traffic, leading to delays and dropped packets. Explain the concept of congestion control and describe at least two mechanisms used to manage network congestion.

Congestion occurs when the **load (number of packets) offered to the network exceeds its capacity** - router buffers/queues fill up, causing long **delays**, **packet loss** and retransmissions that make it even worse (congestion collapse). **Congestion control** is the set of techniques that **prevent congestion or remove it** once it happens, keeping the load below capacity.

A. Open-loop (prevention) policies

- **Retransmission policy** - good timers so senders don't retransmit too early.
- **Window policy** - Selective Repeat instead of Go-Back-N (fewer duplicates).
- **Acknowledgement policy** - cumulative/delayed ACKs reduce extra traffic.
- **Discard policy** - routers drop less important packets first (e.g. in audio).
- **Admission control** - a new flow/virtual circuit is refused if the network cannot support it (used in QoS, ATM).
- **Traffic shaping** - **Leaky bucket**: packets leave at a constant rate regardless of bursts (smooths traffic; overflow dropped). **Token bucket**: tokens added at a fixed rate; a packet needs a token to be sent, allowing controlled bursts.

B. Closed-loop (removal) mechanisms

- **Backpressure** - a congested node stops accepting packets from upstream nodes, which in turn slow down, back to the source.
- **Choke packet** - the congested router sends a special packet directly to the source telling it to slow down (e.g. ICMP source quench).
- **Implicit signalling** - the source infers congestion from missing ACKs / delay (TCP).
- **Explicit signalling** - routers mark packets (**ECN** bits in IP header, Frame Relay FECN/BECN) to tell sender/receiver.
- **Random Early Detection (RED)** - router drops/marks packets randomly before the queue is full so TCP senders slow down early.

C. TCP congestion control

- **Slow start** - congestion window (cwnd) starts at 1 MSS and **doubles each RTT** (exponential) until the threshold (ssthresh).
- **Congestion avoidance (AIMD)** - after the threshold cwnd grows by 1 MSS per RTT (additive increase).
- **On loss**: timeout -> ssthresh = cwnd/2, cwnd = 1 (back to slow start); 3 duplicate ACKs -> **fast retransmit & fast recovery** (cwnd halved - multiplicative decrease).

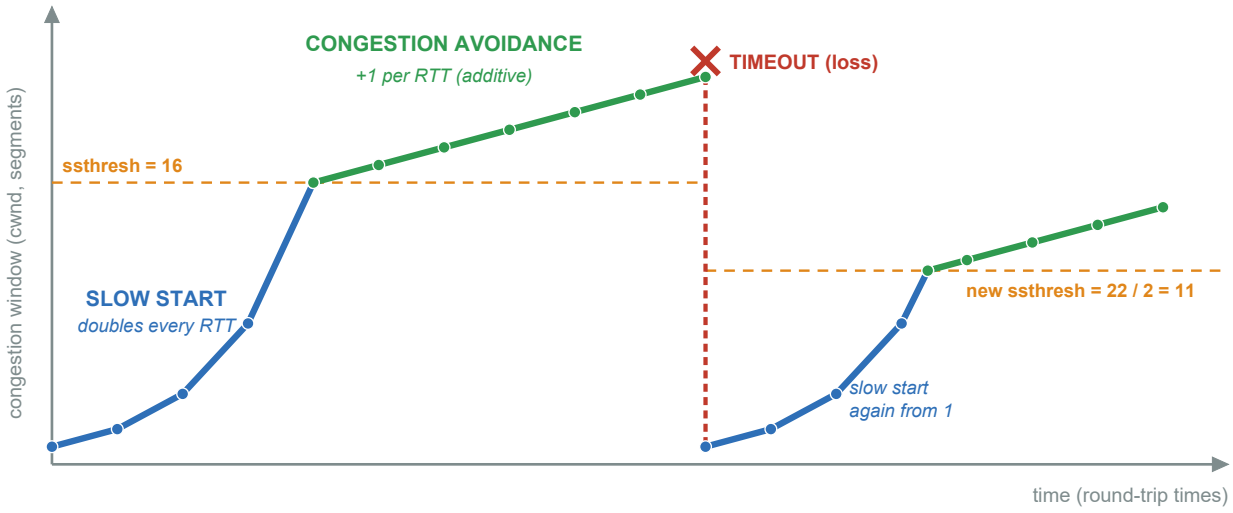


Figure: TCP congestion control - slow start, AIMD and timeout

EXAM TIP: For 10 marks: define congestion + causes (2), define congestion control (1), then explain 2-3 mechanisms in detail (leaky/token bucket, choke packet, TCP slow start) with a diagram (6-7).

6. Quality of Service (QoS)

EXAM QUESTION 2023 Q3d **[4]**
The Internet is often described as a 'best effort network'. Explain what is meant by this term.

In a **best-effort network** the network **tries its best to deliver every packet but gives no guarantees**: no guarantee of delivery, of delivery time (delay), of bandwidth, of order or of no duplication. All packets are treated **equally** (first-come-first-served); when routers are congested packets are simply **dropped**. IP is best-effort and connectionless - reliability, if needed, is provided by the end hosts (TCP). Advantage: simple, cheap and scalable; disadvantage: unpredictable performance for real-time applications, which is why QoS mechanisms (DiffServ, IntServ/RSVP, MPLS) were added.

EXAM QUESTION 2023 Q3e **[6]**
List three QoS parameters that are often measured to characterise the behaviour of a network or network connection.

- **Bandwidth / throughput** - the amount of data successfully transferred per second (bps).
- **Latency (delay)** - time for a packet to travel from source to destination (propagation + transmission + queuing + processing).
- **Jitter** - **variation in delay** between packets of the same flow; harms real-time audio/video.
- **Packet loss** - percentage of packets that never arrive (caused by congestion/errors).
- (Also reliability/error rate and availability.)

EXAM QUESTION 2023 Q2e **[5]**
Discuss the quality-of-service requirements of a VoIP application and how they differ from those of a video-based application.

Parameter	VoIP (voice)	Video application
Bandwidth	**Low** - ~30-100 kbps per call (G.711 ~87 kbps, G.729 ~30 kbps)	**High** - 1-25 Mbps (HD/4K), bursty
Latency	Very sensitive: one-way delay < **150 ms**	Streaming (YouTube/Netflix) tolerant - **buffering** hides delay (seconds); video conferencing needs < 150-200 ms like voice
Jitter	Must be low (< **30 ms**); jitter buffers used	Streaming tolerant (buffer); conferencing sensitive
Packet loss	< **1%**; lost packets are not retransmitted (UDP/RTP)	Sensitive (< 0.1-1%) - loss causes pixelation/freezing
Traffic pattern	Smooth, constant small packets, interactive two-way	Large packets, variable rate, mostly one-way (streaming)
QoS treatment	Highest priority (DSCP EF 46), priority/low-latency queuing	Lower priority (AF41 conferencing, AF31 streaming)

EXAM QUESTION ISA2024 Q4d **[7]**
A large enterprise network is experiencing slow data transfer rates, high latency and frequent packet losses. Describe network performance optimisation and how you would achieve it to improve efficiency, speed and reliability.

Network performance optimisation = monitoring, analysing and tuning the network (hardware, configuration and traffic) so that it delivers the maximum throughput with the minimum latency, jitter and loss for business applications.

1. **Monitor & baseline** - use SNMP/NetFlow tools (PRTG, Wireshark, SolarWinds) to find bottlenecks, top talkers, error counters, overloaded links.
2. **Upgrade bandwidth & hardware** - faster links (1 -> 10 Gbps), replace hubs with switches, upgrade old routers/NICs and faulty cables (fixes CRC errors & loss).
3. **Segment the network** - VLANs/subnetting to reduce broadcast traffic and collision domains.
4. **Implement QoS** - classify and prioritise critical traffic (VoIP, ERP) over bulk downloads; traffic shaping/policing.
5. **Load balancing & redundancy** - multiple links (EtherChannel), redundant routers (HSRP), dynamic routing (OSPF/EIGRP) for fast failover -> reliability.
6. **Caching, CDN & WAN optimisation** - proxy caches, compression, deduplication; move apps closer to users.
7. **Security hygiene** - block malware/broadcast storms, unauthorised streaming with firewall policies; keep firmware updated; document and review regularly.

7. Software-Defined Networking (SDN)

EXAM QUESTION ISA2024 Q4b **[5]**
Elaborate on software-defined networking (SDN).

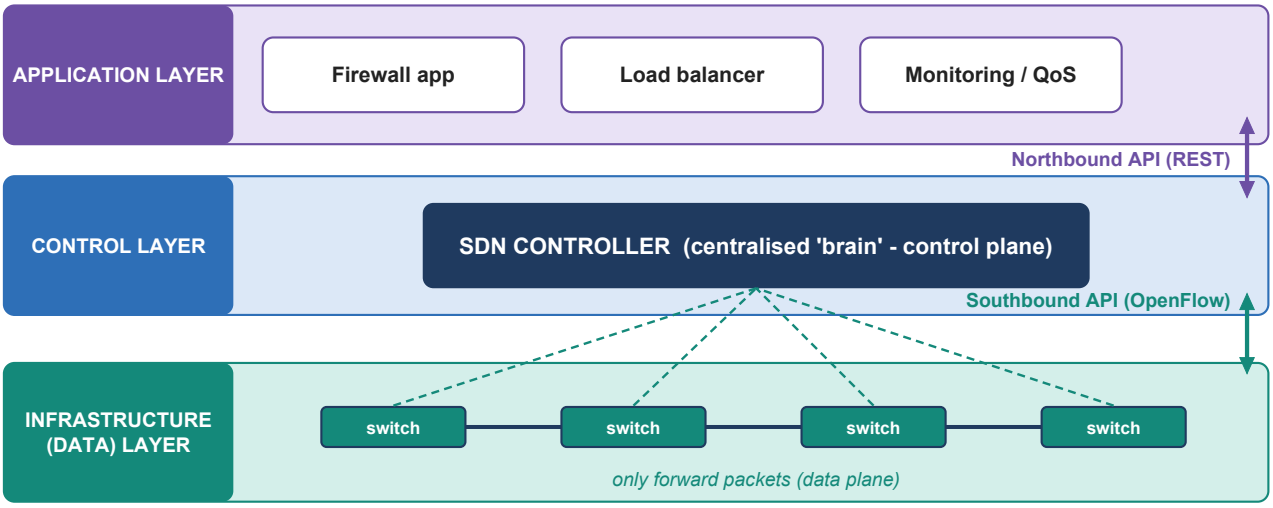


Figure: SDN architecture

- SDN is an architecture that **separates the control plane (decides where traffic goes) from the data plane (forwards the packets)**.
- Control is **centralised in software (an SDN controller)**, e.g. OpenDaylight, ONOS, Cisco ACI; devices become simple forwarding elements.
- The controller programs devices through **southbound APIs** (e.g. **OpenFlow**); applications ask the controller for services through **northbound APIs**.
- **Benefits:** network is **programmable** and automated; central management and visibility; faster provisioning; vendor independence; easier to apply policies/QoS/security network-wide; supports cloud and virtualisation.
- **Challenges:** the controller is a single point of failure/attack, scalability, skills needed.

8. Hierarchical structure of the Internet (ISPs)

EXAM QUESTION 2018 Q4c **[6]**
Draw and explain the hierarchical structure of the Internet (ISPs).

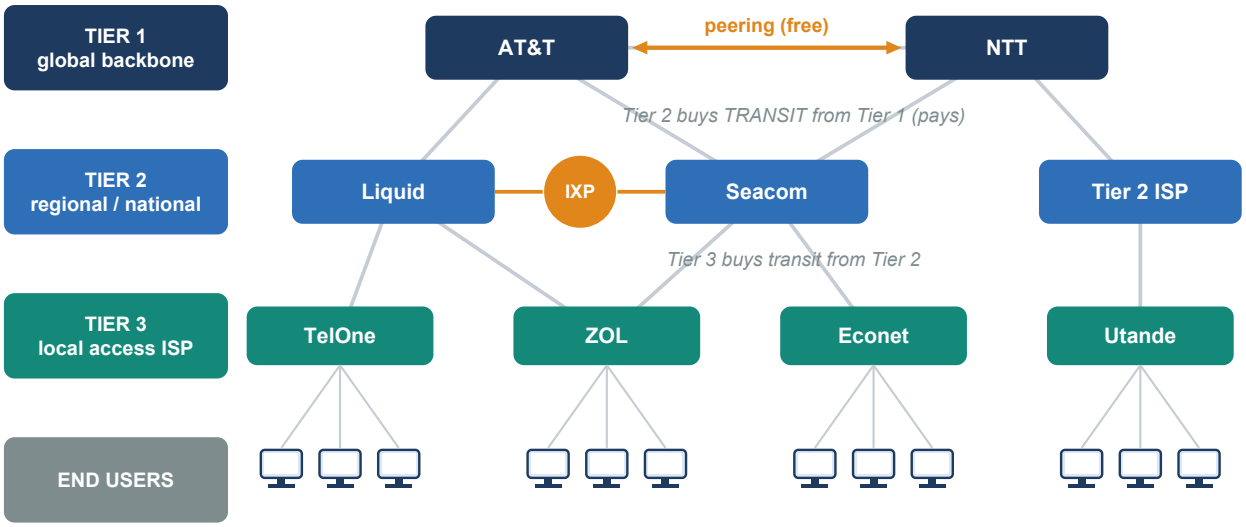


Figure: hierarchical structure of the Internet

- **Tier 1 ISPs** - global backbone networks that reach the whole Internet **without paying for transit**; they **peer** with each other (settlement-free). E.g. AT&T, NTT, Tata, Lumen.
- **Tier 2 ISPs** - regional/national providers; **buy transit from Tier 1** and peer with other Tier 2s at **IXPs (Internet Exchange Points)**. E.g. Liquid Intelligent Technologies, Seacom.
- **Tier 3 ISPs** - local/access ISPs that buy transit from Tier 2 and connect end users (homes, businesses). E.g. ZOL, TelOne, Econet, Utande.
- Connections happen at **POPs (points of presence)** and **IXPs**; content providers (Google, Netflix) also connect directly. Traffic flows up the hierarchy and back down.

9. Service Level Agreement (SLA)

> **EXAM QUESTION** 2023 Q5d **[1 + 2]**
> Telcos define the services they offer within an SLA. i) Explain the purpose of an SLA. ii) Give two examples of what it might contain.

i) An SLA is a **formal contract between a service provider and a customer that defines the level and quality of service** to be delivered, how it is measured and what happens (penalties/credits) if it is not met.

ii) Contents: **guaranteed availability/uptime** (e.g. 99.9%); **bandwidth/throughput** guaranteed; **maximum latency, jitter and packet loss**; **response and repair (fault resolution) times**; support hours/contact; **penalties/service credits**; security and data protection responsibilities; reporting/monitoring.

10. WAN protocols & point-to-point networks

> **EXAM QUESTION** SE2024 Q1c **[2]**
> Explain two WAN protocols.

- **PPP (Point-to-Point Protocol)** - Layer 2 protocol for direct links between two routers/nodes (serial, DSL - PPPoE); provides framing, authentication (**PAP/CHAP**), error detection and multi-protocol support via LCP/NCP.
- **HDLC (High-level Data Link Control)** - bit-oriented synchronous data-link protocol for point-to-point/multipoint links; Cisco default on serial interfaces; uses flags 01111110 and bit stuffing.
- **Frame Relay** - packet-switched WAN using virtual circuits (PVCs, DLCI numbers); cheaper than leased lines (now legacy).
- **MPLS (Multiprotocol Label Switching)** - routers forward packets based on short **labels** instead of IP lookups; fast, supports QoS and VPNs - used by telcos today.
- Others: **ATM** (fixed 53-byte cells), **X.25** (old, error-checked packet switching), ISDN, leased lines (E1/T1).

> **REMEMBER:** Routing table sources: Connected, Static, Dynamic. DV = Bellman-Ford, periodic full tables, hop count (RIP). LS = Dijkstra, LSAs on change, cost (OSPF). EIGRP = DUAL, successor & feasible successor, bandwidth + delay. QoS = bandwidth, latency, jitter, loss.